

Informal Statement (IF)

Consider a prime p of the form $4t+3$. Show that a is a primitive root modulo p iff $-a$ has order $(p-1)/2$.

Q_1 : A prime number p is ..., formally expressed in Lean as $p : \dots$

Q_2 : A primitive root modulo p is ..., in Lean using ..., which asserts that ...

Q_3 : The negation $-a$ in modular arithmetic... ring structure of $\mathbb{ZMod} p$

Q_4 : ... $-a$ has order $(p-1)/2$ means ... expressed as $\text{orderOf}(-a : (\mathbb{ZMod} p)) \dots$

Q_5 : The biconditional statement ' a is a primitive root modulo ... $a \leftrightarrow \text{orderOf}(-a : (\mathbb{ZMod} p)^x) = \dots$

Q_6 : The multiplicative order of an element a ..., using $\text{orderOf} a$ in $(\mathbb{ZMod} p)$...



① Decompose



PNat.Prime



primitiveRoots



$\mathbb{ZMod}$



IsPrimitiveRoot



orderOf

② Retrieve

Theorem 1 ... $(h : \text{IsPrimitiveRoot } \mu n) \dots$
 $(\text{Int.castRingHom } (\mathbb{ZMod} p)) \dots$

Theorem 2 ... $\{a : \mathbb{ZMod} p\} \dots (ha : a \neq 0) : \text{orderOf } a \mid p-1$

Theorem 3 ... $\{p : \mathbb{N}^+\} : \text{PNat.Prime } p \rightarrow 1 < p \dots$

③ Illustrate

{Formalization Instruction}

{Premise Set}



{Theorem Set}



{Informal Statement}



④ Formalize Theorems

Formal Statement (FS)

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theorem Ireland-Rosen.exercise_4_5 {p t : ℕ} (hp0 : p.Prime) (hp1 : p = 4*t + 3) (a : ℤMod p) : IsPrimitiveRoot a
  ↔ ((-a) ^ ((p-1)/2) = 1 ∧ ∀ (k : ℕ), k < (p-1)/2 → (-a)^k ≠ 1) := sorry
```